D5: Research Note - How Retail Loan Approval Works

Introduction
For this assignment, I focused on the **Personal Loan** product for a mid-sized Indian bank. Before building the demo, I researched how the retail lending process works. The goal was to understand where the delays happen and how AI could help.

The Current Process & Roles
Typically, a personal loan goes through five main stages:

1. **Application:** The customer submits details and documents (KYC, salary slips, bank statements).
2. **Document Check:** A back-office team manually verifies if all documents are present and readable.
3. **Credit Assessment:** The Credit Officer pulls the CIBIL score, calculates the FOIR (Fixed Obligation to Income Ratio), and checks internal policies.
4. **Underwriting & Sanction:** A senior officer reviews the recommendation and approves or rejects it.
5. **Disbursement:** The loan agreement is signed, and money is transferred.

Pain Points in Today's System
Based on standard banking processes and public RBI guidelines, the biggest issues are:

- **High Turnaround Time (TAT):** It usually takes 5 to 9 days to approve a loan because everything is manual.
- **Inconsistent Decisions:** Different credit officers might interpret policy rules differently.
- **Data Silos:** Core banking data, CIBIL scores, and physical documents are not connected. A human must pull them together manually.

Conclusion
A modern solution needs a Lakehouse to bring data together and an AI Agent to do the heavy lifting (calculating math, reading documents), while keeping a human officer in control for the final compliance checks.

Sources Consulted:

1. Public RBI Guidelines on Retail Lending and KYC norms.
2. Industry reports on Turnaround Time (TAT) in Indian Retail Banking.

D6: AI Usage Log

Project: Agentic AI Loan Approval Demo
Tools Used: ChatGPT (for writing/coding help),DeepSeek AI, Groq API (for the AI Agent).

How I Used AI:
As a beginner, I used AI as my pair-programmer and research assistant to build this. Here is how it helped:

1. **Research (D5):** I used ChatGPT to draft the research note. I prompted it to outline the personal loan stages, roles, and TAT pain points in Indian banks.
2. **Synthetic Data Generation:** Instead of typing 200 applicant records, I asked AI to generate a JSON dataset of realistic Indian names, random loan amounts, and CIBIL scores (with 15% missing documents) to simulate real-world data.
3. **UI Design:** I used AI to generate the CSS and HTML for the "Retail Lending Workspace" UI, so it looked like a real banking dashboard.
4. **The Agentic Logic (The Core):** I used AI to help me write the JavaScript for the Groq API. It showed me how to define the calculate_foir tool and how to handle the "Function Calling" loop (where the AI asks for math, JS calculates it, and sends it back).

What Worked & What I Corrected:

- **Worked:** AI was amazing at generating boilerplate code and explaining how API endpoints work. It saved me at least **15 hours** of manual coding and reading documentation.
- **Correction:** Initially, the AI used LaTeX math formatting in its output (like \frac{}{}), which broke the UI. I had to correct the prompt to tell the AI to use plain text for math.
- **Correction:** I also had to troubleshoot the Groq model retirement (switching from llama-3.3 to gpt-oss-120b) which AI helped me identify.

Reflection:
Using AI didn't just help me write code faster, it helped me understand *how* to build an Agentic system without getting stuck on syntax errors. It allowed me to focus on the business logic and the story, which is the most important part of a presales demo.